

ECEN 474/704: Lab 06

Differential Pairs(Simulations on simple Differential Configuration and Cascode Current Mirror Configuration)

Sheena Selvakumar

UIN: 537000654

Due date: 2025-10-31

I. DISCUSSION OF THE LAB AND RESULTS

Include a short write-up (a paragraph or two) summarizing the lab and results. Compare the final designs with your pre-lab designs (as applicable for the lab), and make note of any major changes in your designs and why these changes were made. Regarding the simulation results, briefly describe any trends that are shown from the data or differences between pre- and post-layout simulation results, and attempt to provide an explanation for the trends and/or differences. Additionally, consider some of the following questions: were the results as you expected? Why or why not? What, if any, design trade-offs are demonstrated by the work in the given lab? If you were unable to meet any specifications given in the lab, what do you believe was the primary cause and what could be some potential solutions for the future?

The above questions are not an exhaustive list of what could be discussed and may not be relevant for all labs, but rather they serve as a starting point for you to demonstrate your understanding of the lab material.

In this lab, with the given high level design constraints we tried to design simple differential pair and current mirror cascode differential pair, we tried to calculate and analyse:

1. SLEW RATE
2. GBW
3. DIFFERENTIAL AND COMMON MODE
4. POLE FREQUENCY

Even though the w/l sizing and matching is done in prelab, it has to be slightly modified due to the hit and trial error for the simulation to meet the device specification in the cadence software

I honestly faced issue while designing the current mirror cascode differential pair, I am still ambiguous that my differential and common mode gains are low, but my CMRR meets the spec.

II. SUMMARY OF RESULTS

Summarize the simulation results from the lab in one or two tables, like what is shown below.

Table 1 for the current mirror cascode differential pair
Example summary table of pre- and post-layout results

Parameter	Gain	GBW	Slew Rate	CMRR	Common-mode Input Range	Pole freq
Pre-layout	39.4747 db	8.154MHz	15.23V/us	37.9091	0.715	426.846kHz
Post-layout	39.4747 db	8.154MHz	15.23 V/us	37.5965	0.6	426.846kHz

I_{bias}=180u

w/l for all transistors: (ALL 4 fingers with 3 dummies with)

M1,M2	670n/finger	270n(length)
M3,M4	1.48u/finger	270n
M5,M6	1.96u/finger	270n

Table 2 for the simple differential pair
Example summary table of pre- and post-layout results
 $I_{bias}=95\mu$

Parameter	Gain	GBW	Slew Rate	CMRR	Common-mode Input Range	Pole freq
Pre-layout	-11.5 db	184.626	999.7V/ms	65.0162	0.06	707.9kHz
Post-layout	-6.96db	199.251	999.7V/ms	67.0937	0.06	701.062kHz

w/l for all transistors:2 fingers each

M9,M10	500n	270n
M7,M8	500n	270n
M1,M2	5.5 u	270n
M3,M4	8.5 u	270n
M5,M6	8.5 u	270n

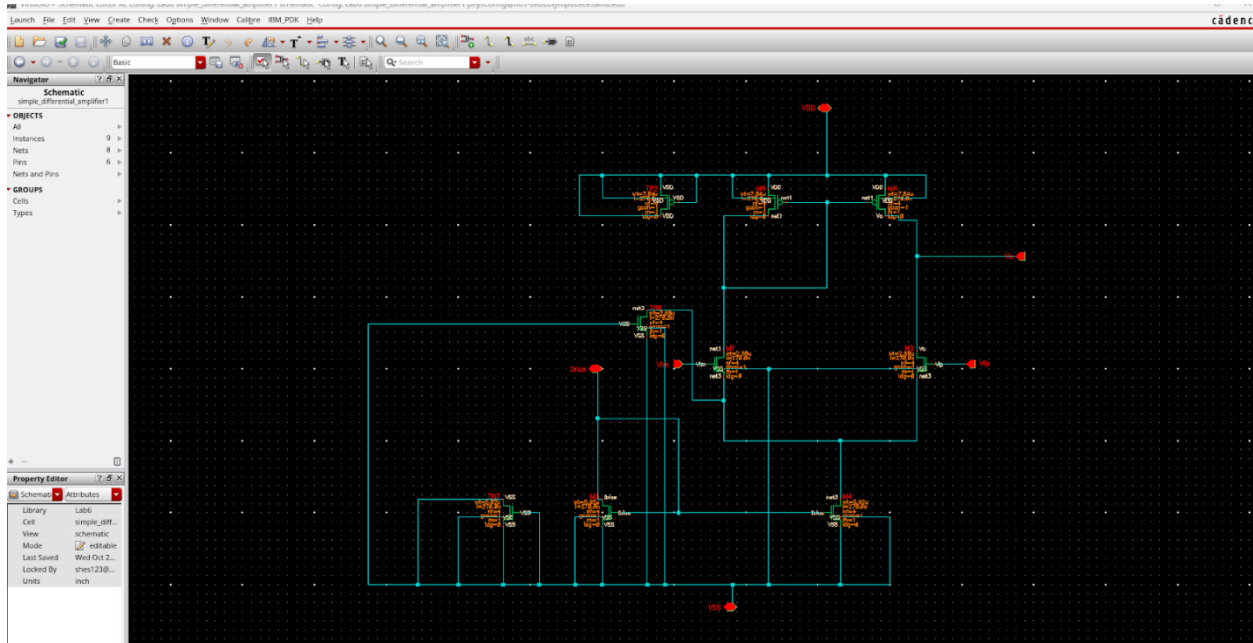
$V_{bn}=75\text{Mv}$ and $V_{bp}=175\text{ mV}$

III. SUB-CIRCUIT AND TEST BENCH SCHEMATICS

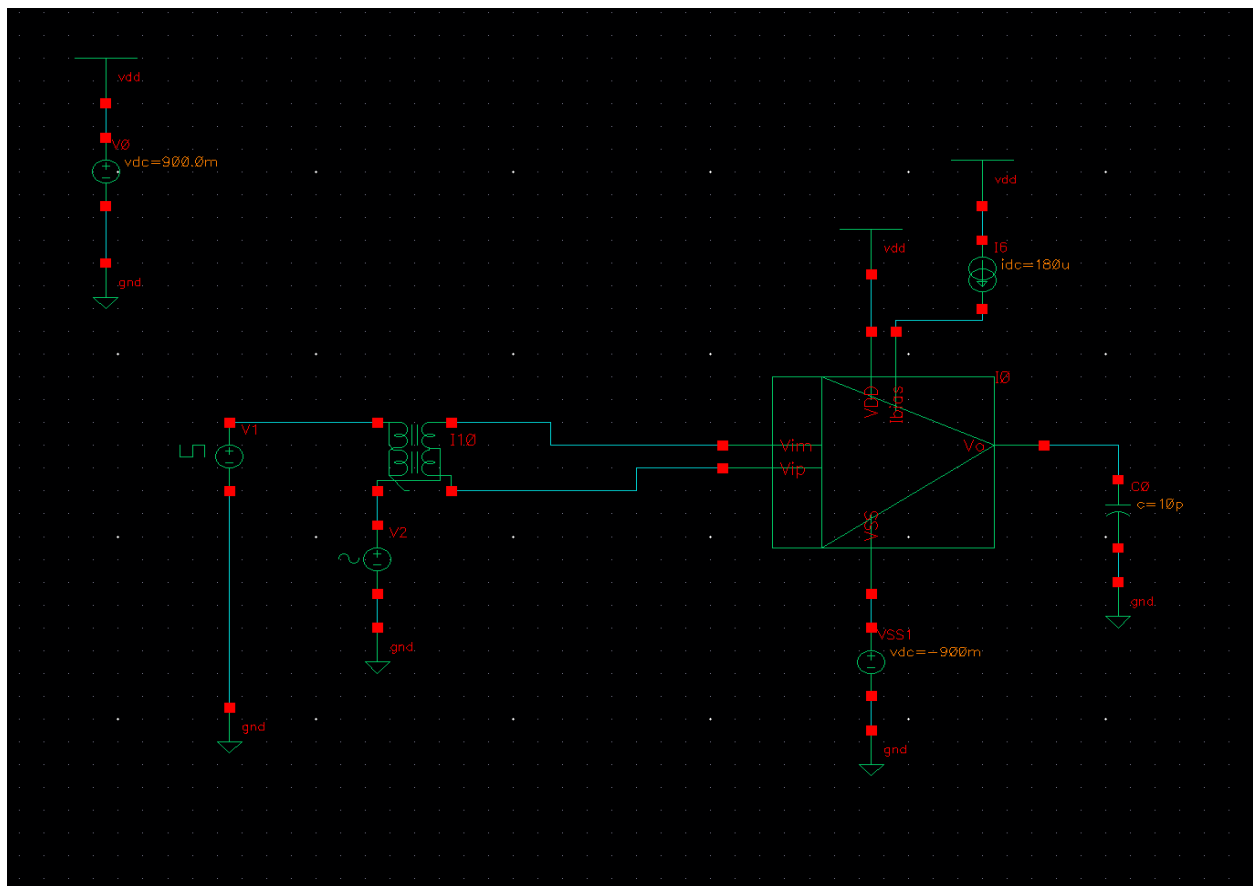
Schematics of the sub-circuit(s) being designed in the lab should be included here. In addition, include images of the test bench schematics. Important signals in the test bench schematic should be labeled to provide clear correspondence between the signals in the schematics and the signals that are plotted in the simulation results shown later in the report.

1.for simple differential pair:

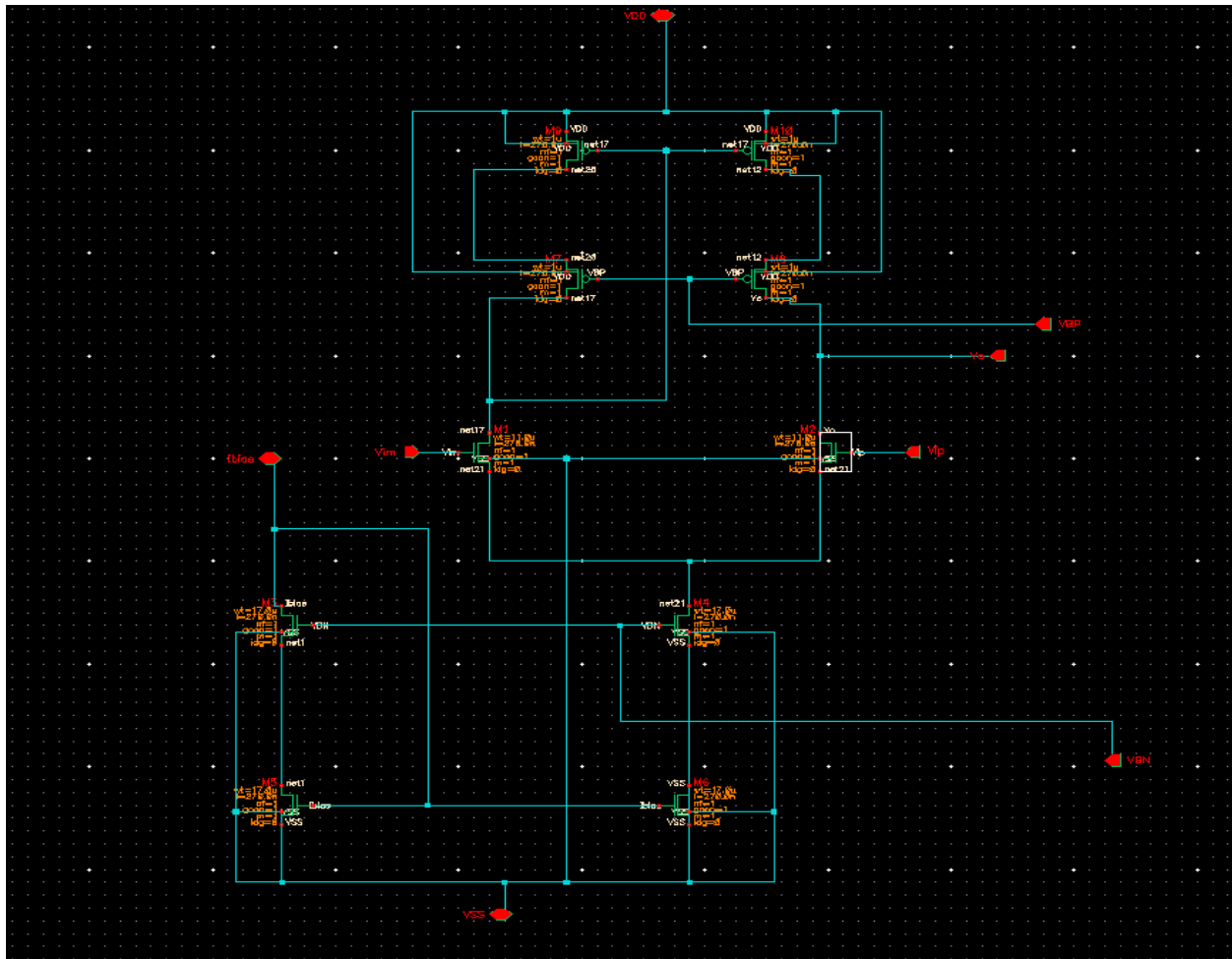
Schematic



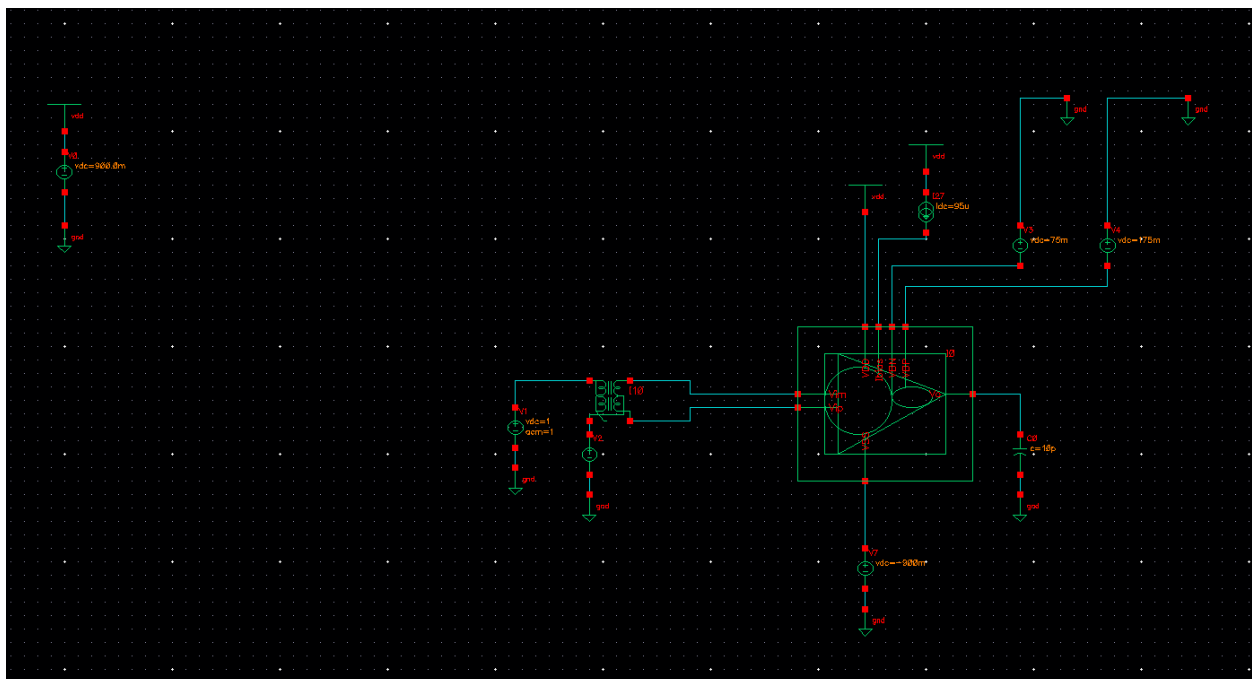
Testbench:



2.for current mirror cascode differential pair:
1.Schematic



2. Test bench:



IV. LAYOUT(S) AND FLOOR PLAN(S)

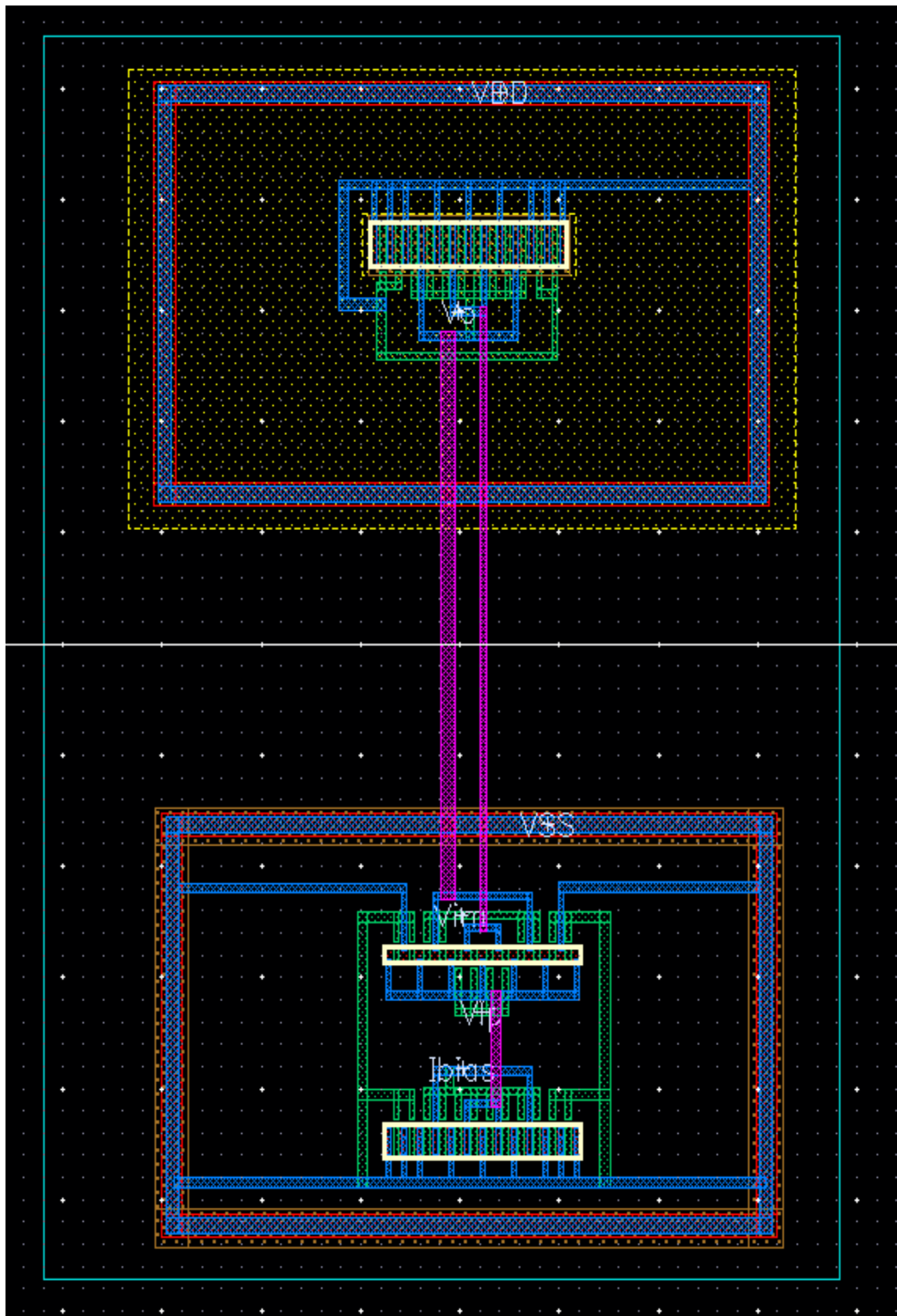
Include diagrams of the floor plans used to lay out major circuit components. Typically, this will just include transistor floor plans (routing does not need to be included in the diagrams). A table can be used to show the layout arrangement for the transistors or the floor plan can be drawn in another software tool. Clearly label the devices in the floor plan and make sure there is clear correspondence with the devices in the schematics, and include dummy devices in the floor plan diagram (dummy devices are often denoted with an asterisk, *).

This section will also include images of the layout(s) of each circuit constructed for the lab.

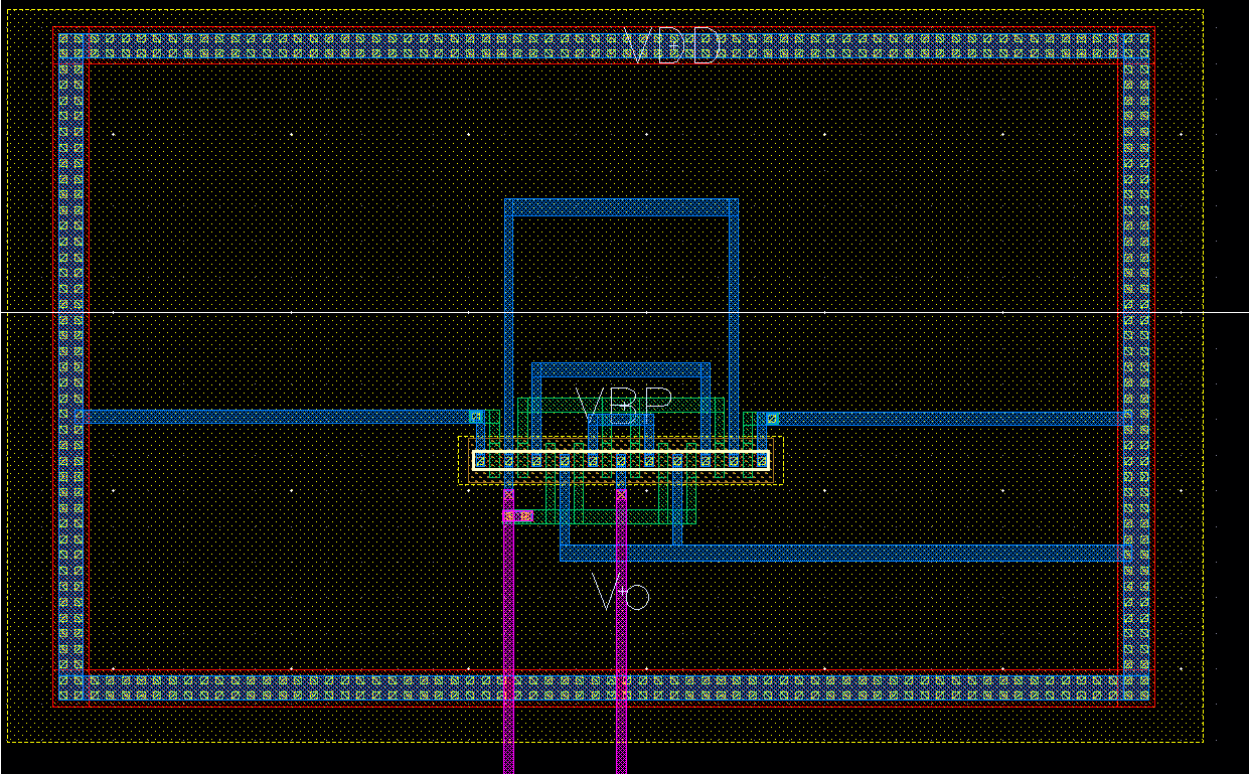
Floor plan of Basic Differential Pair:

FLOOR PLAN 1

D D M5 M5 M6 M6 M6 M6 M5 M5 D D
D D M1 M1 M2 M2 M2 M2 M1 M1 D D
D D M3 M3 M4 M4 M4 M4M3 M3 D D



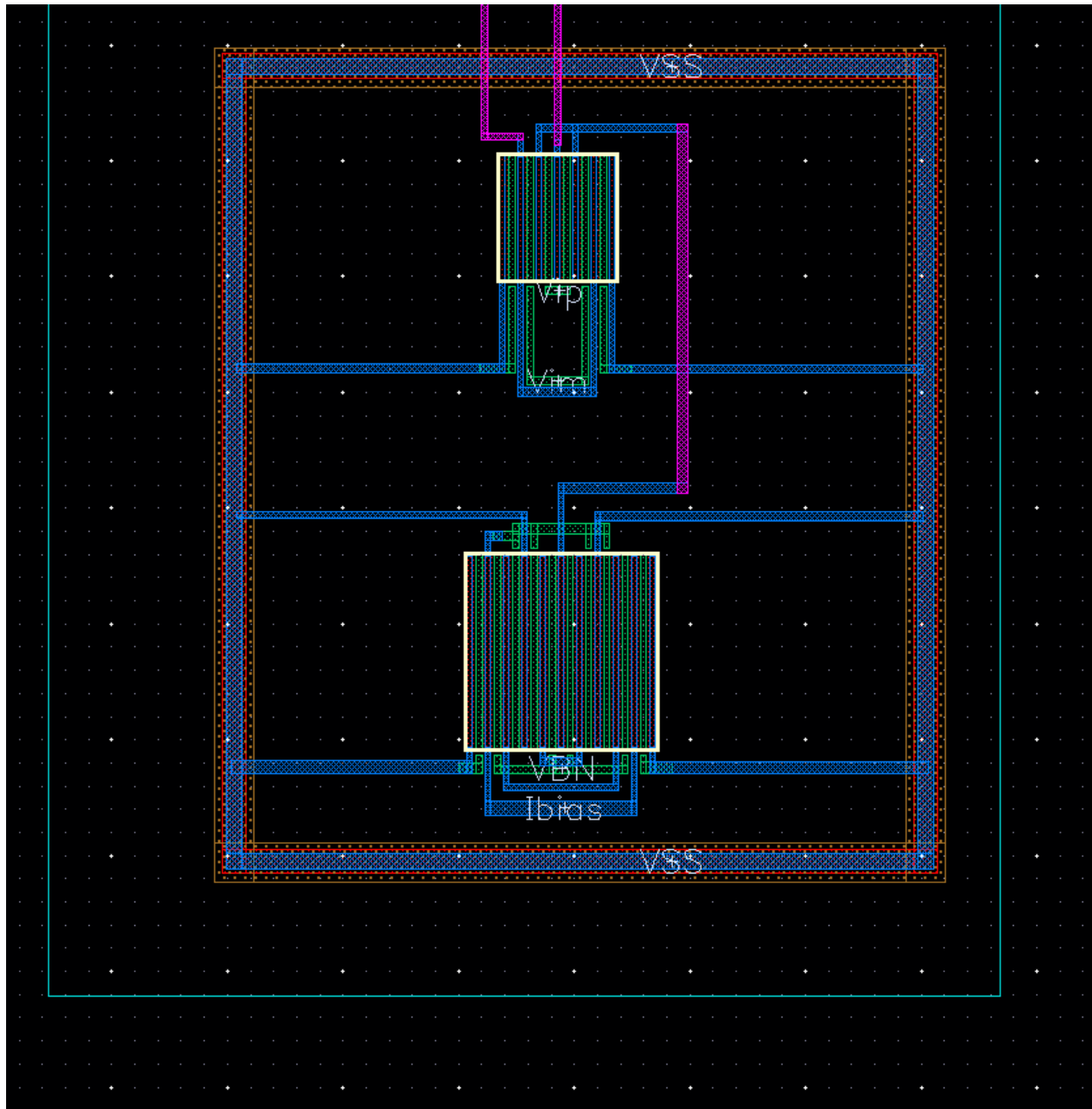
Closer look pmos:



Closer

look

nmos:



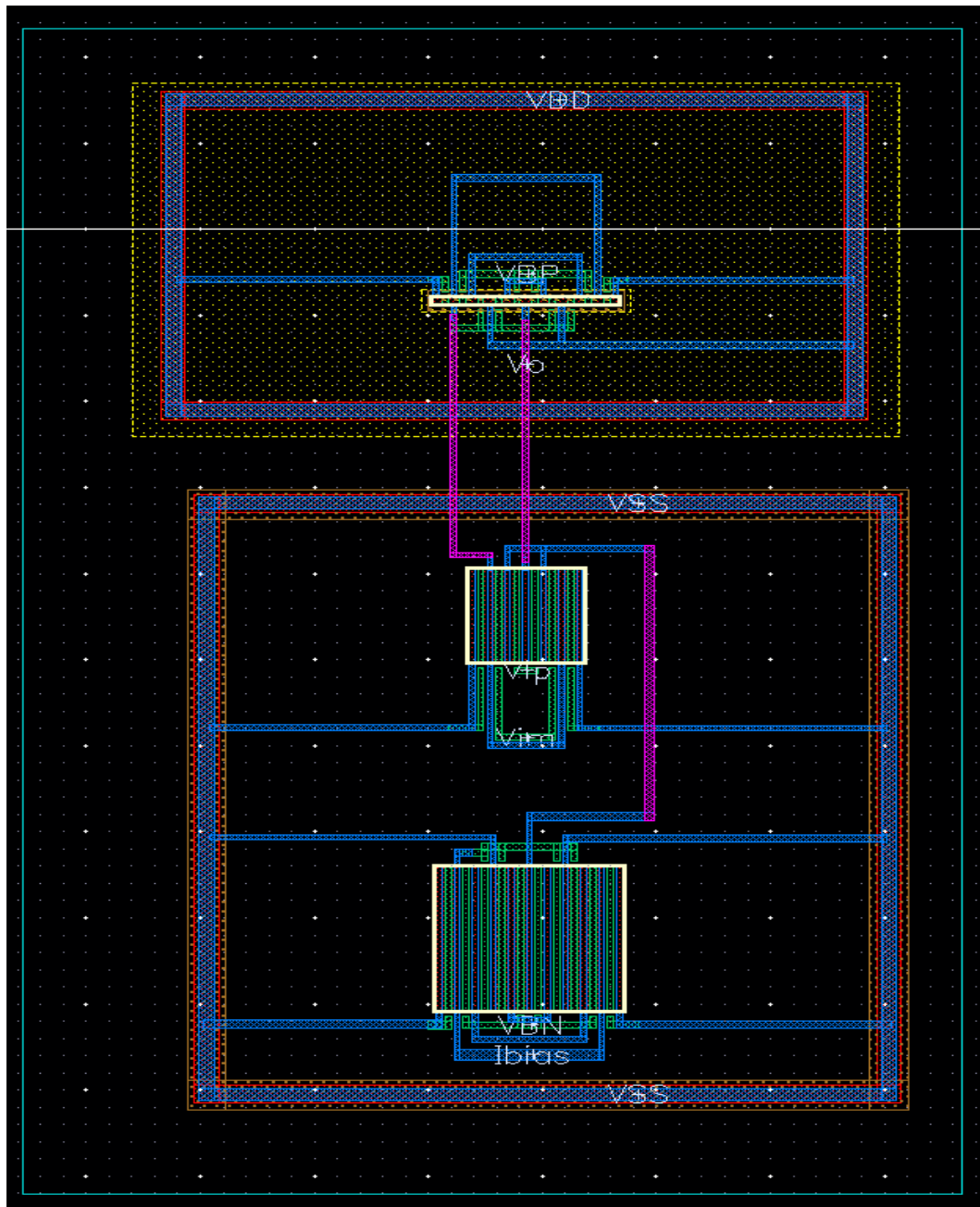
Floor plan of Cascode Current Mirror:

FLOOR PLAN 2

D M7 M9 M10 M8 M8 M10 M9 M7 D

D M1 M2 M2 M1 D

D M3 M5 M6 M4 M4 M6 M5 M3 D



V. DRC AND LVS RESULTS

This section should include the DRC RVE window results for each layout cell (with the results filtered to “show unresolved” to show only the remaining DRC violations), and the cell name should be visible at the top of the DRC RVE window. The LVS report summary image should also be included for each cell with your NetID, the time stamp, and the smiley face/check mark visible to show that LVS passed.

1. for simple differential pair

Calibre - RVE v2024_2_29.16 : srcdb simple_differential_amplifier1

File View Highlight Tools Window Setup Help

Comparison Results

Layout Cell / Type	Source Cell	Net	Instances	Ports
simple_differential_amplifier1	simple_differential_amplifier1	B1, B5	6L, 6S	6L, 6S

Cell simple_differential_amplifier1 Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

LAYOUT CELL NAME: simple_differential_amplifier1
SOURCE CELL NAME: simple_differential_amplifier1

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	6	6	
Nets:	8	8	
Instances:	24	6	MP (4 pairs)
	12	3	MP (4 pairs)
Total Inst:	36	9	

NUMBERS OF OBJECTS AFTER TRANSFORMATION

	Layout	Source	Component Type
Ports:	6	6	
Nets:	8	8	
Instances:	4	4	MP (4 pairs)
	2	2	MP (4 pairs)

Calibre Interactive - nmLVS v2024_2_29.16 : /disk/amsc/IBM_PDK/cmhv7sf/re/Calibre/runset.lvs "@n01-zeus.olympus.ece.tamu.edu

File Settings Configurations Help

simple_differential_amplifier1.lvs.report

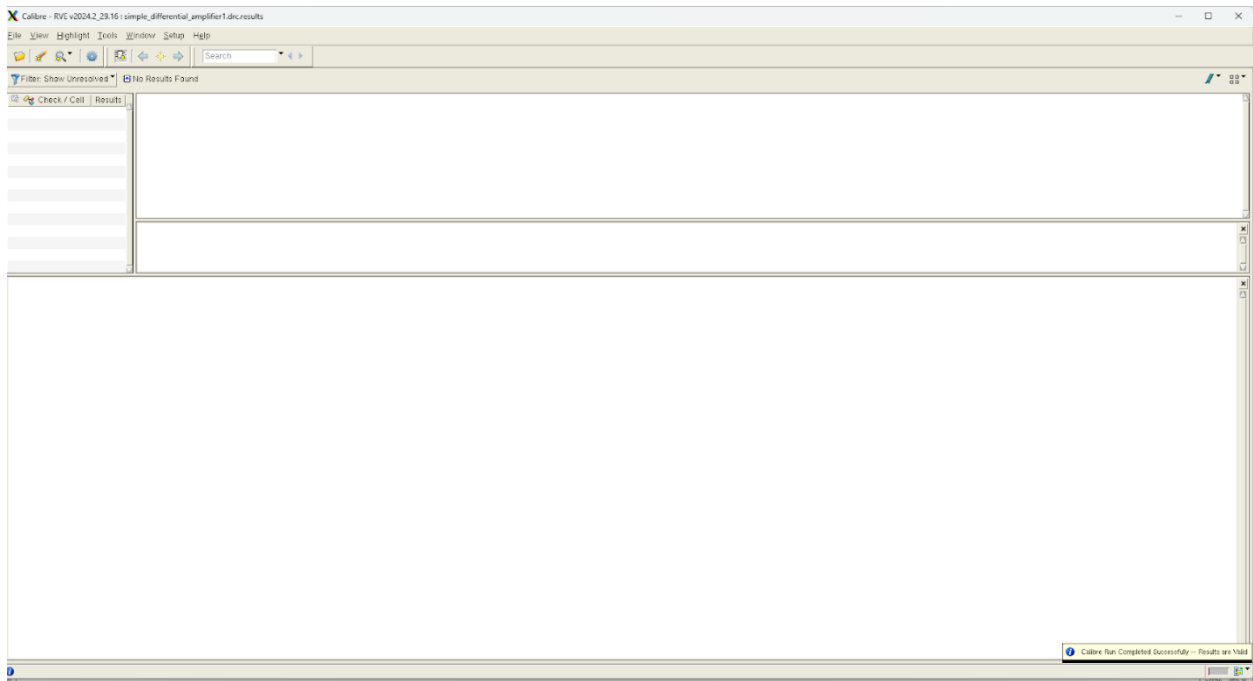
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```
#####  
##  
## CALIBRE SYSTEM ##  
## LVS REPORT ##  
#####  
REPORT FILE NAME: simple_differential_amplifier1.lvs.report  
LAYOUT NAME: simple_differential_amplifier1.sp ('simple_differential_amplifier1')  
SOURCE NAME: simple_differential_amplifier1.src.net ('simple_differential_amplifier1')  
RULE FILE: _cmhv7sf.lvs.cal_  
CREATION TIME: Wed Oct 29 18:41:37 2025  
CURRENT DIRECTORY: /home/grads/s/shes123/CAL  
USER NAME: shes123  
CALIBRE VERSION: v2024_2_29.16 Thu May 2 07:35:42 PDT 2024  
  
OVERALL COMPARISON RESULTS  
  
# ##### - -  
# # * *  
# # CORRECT # |  
# # #  
# #####  
  
=====
```

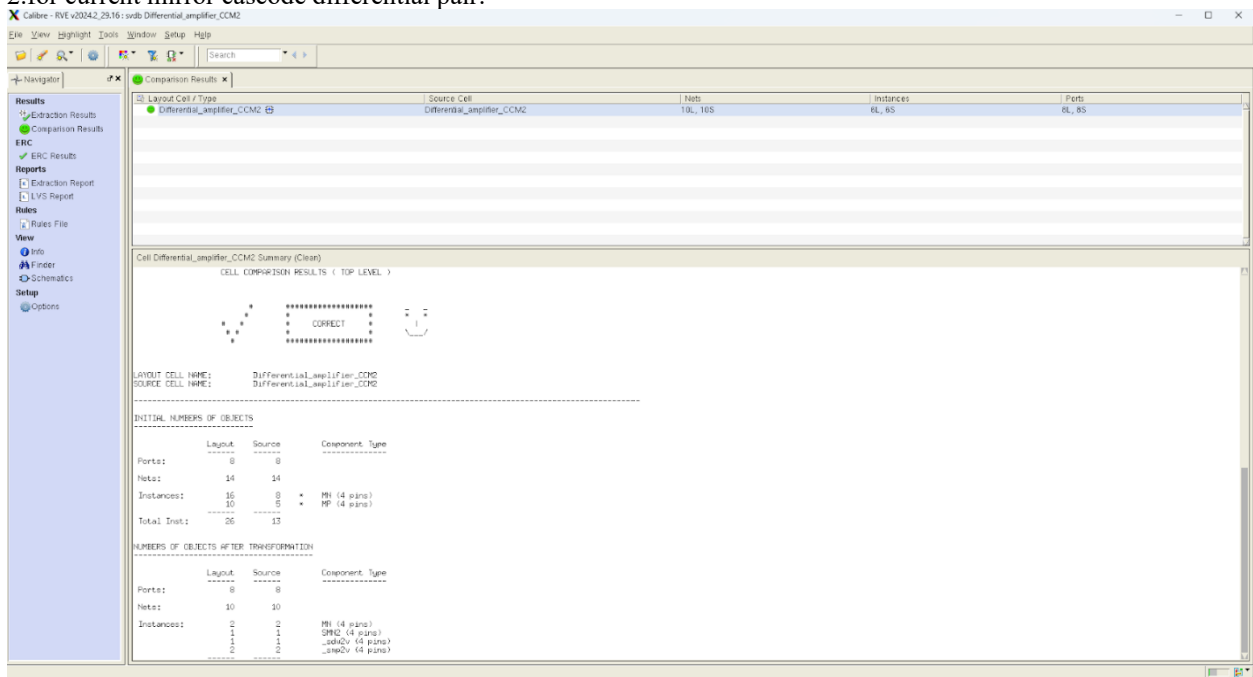
CELL SUMMARY

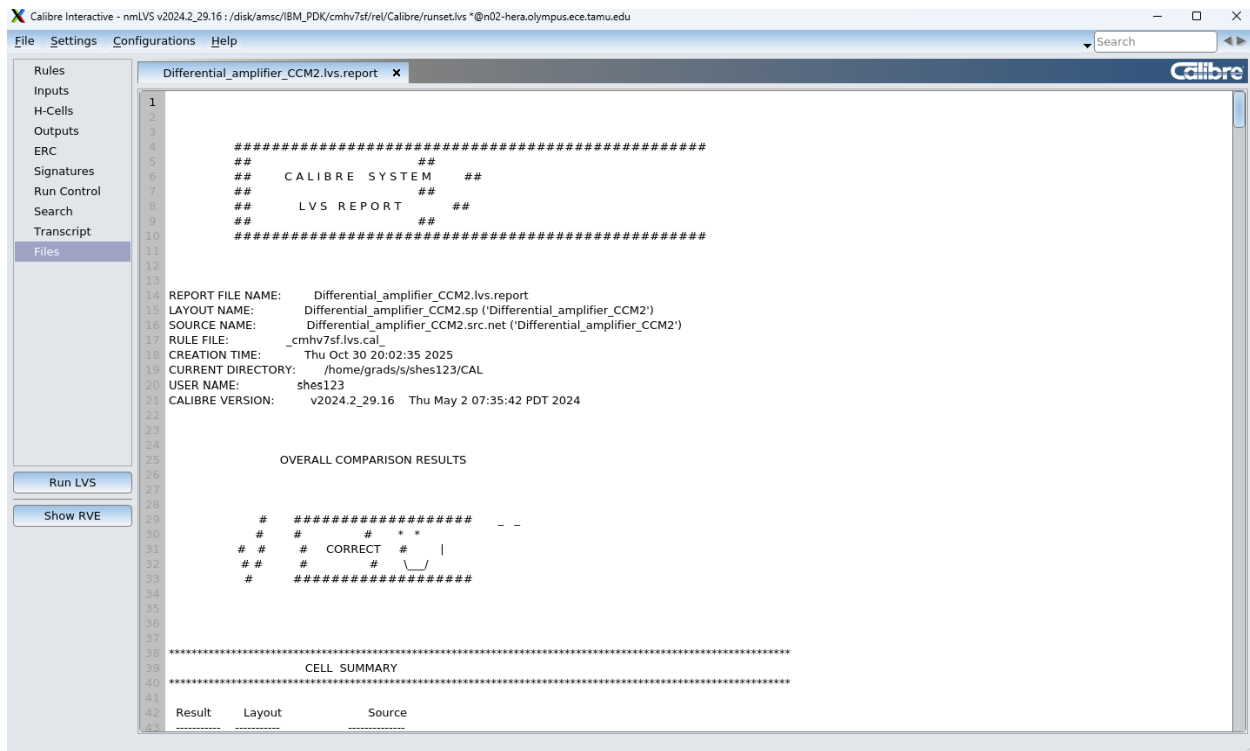
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Result	Layout	Source
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2. for current mirror cascode differential pair:





VI. SIMULATION RESULTS

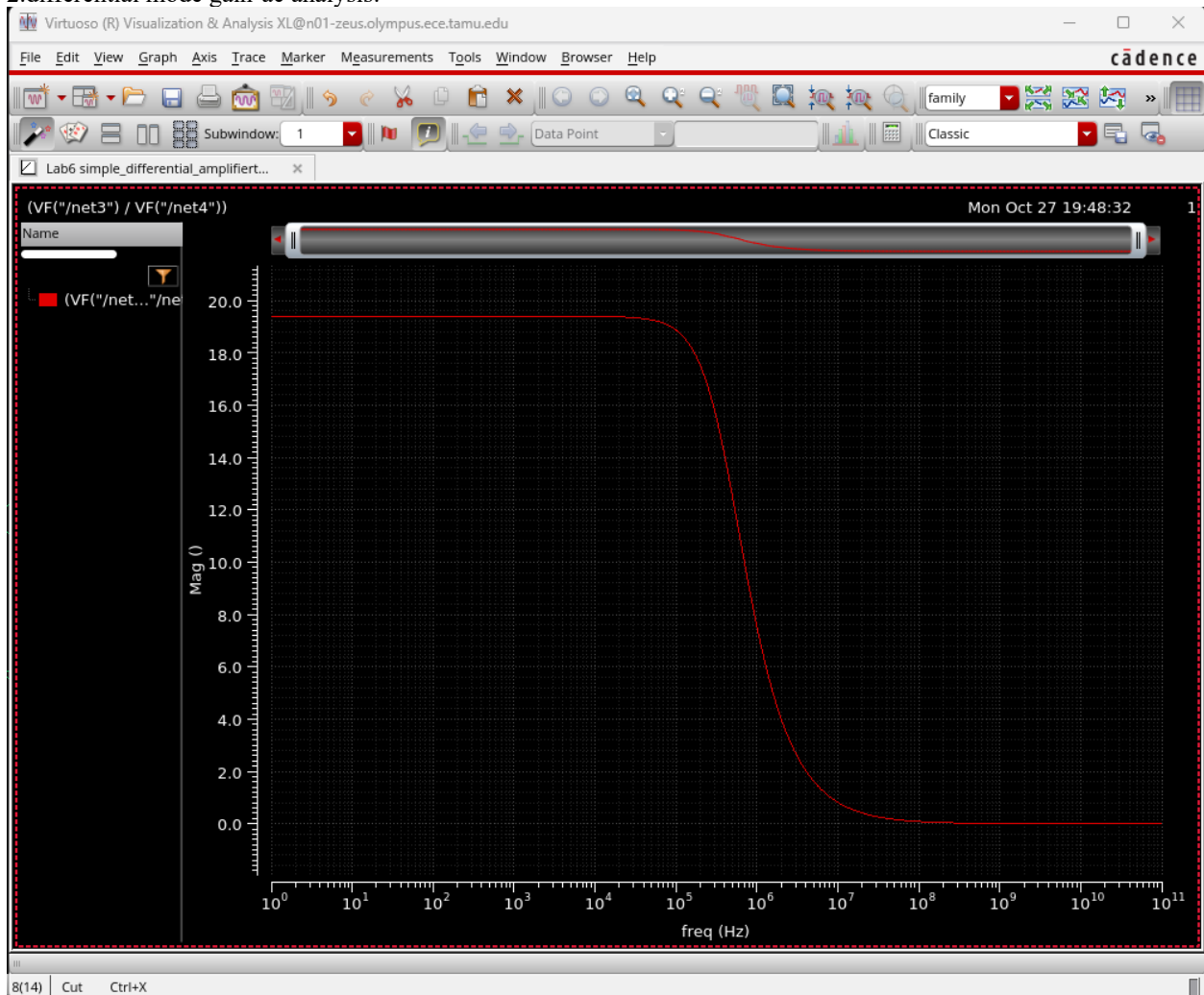
Images of simulation results requested in the lab manual should be included here. Plots should be annotated to show relevant data points which will include either (1) data points that are explicitly requested in the lab manual or (2) data points from which other requested parameters are calculated. Data in the images should also be legible, so scale the images accordingly to make sure the data is easy to read. Post-layout simulations should be plotted in the same window with the pre-layout simulation results in order to make the differences between the two more apparent.

1. for simple differential pair:

1. Common mode input voltage gain:

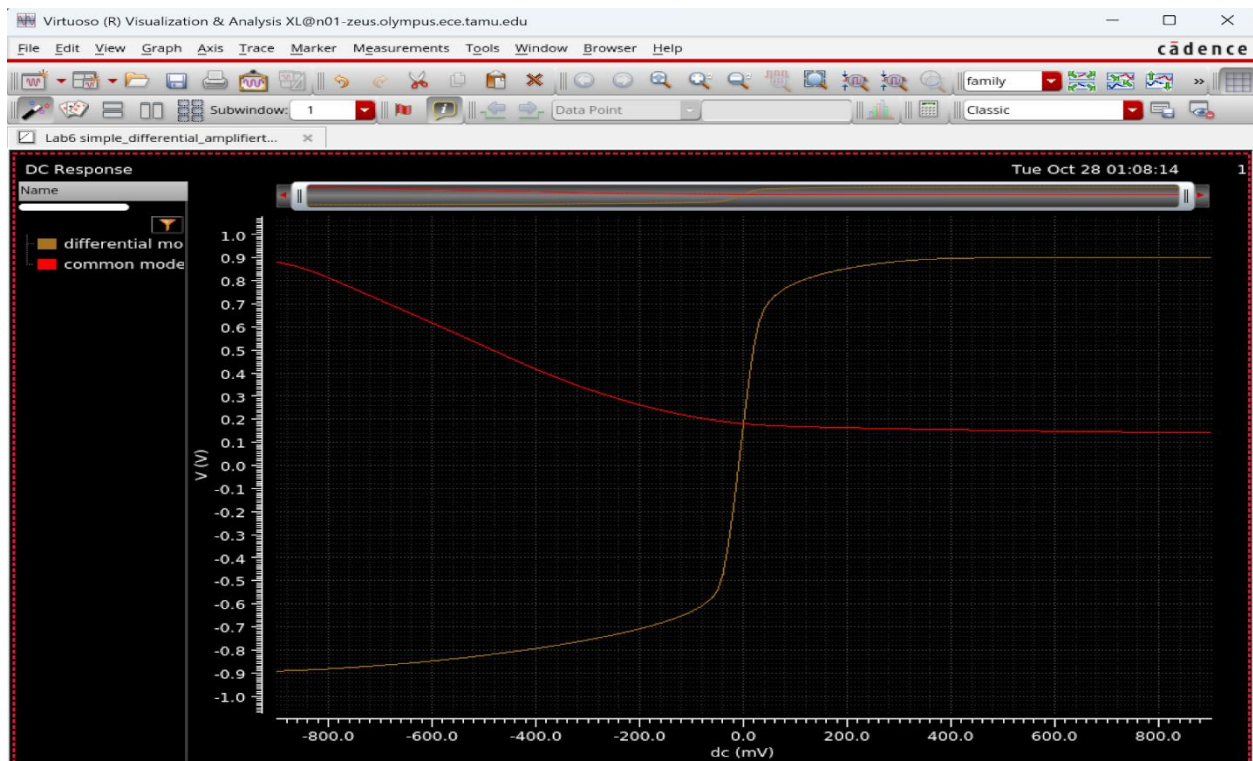


2.differential mode gain-ac analysis:



observing the dc output:

3.Common mode gain:



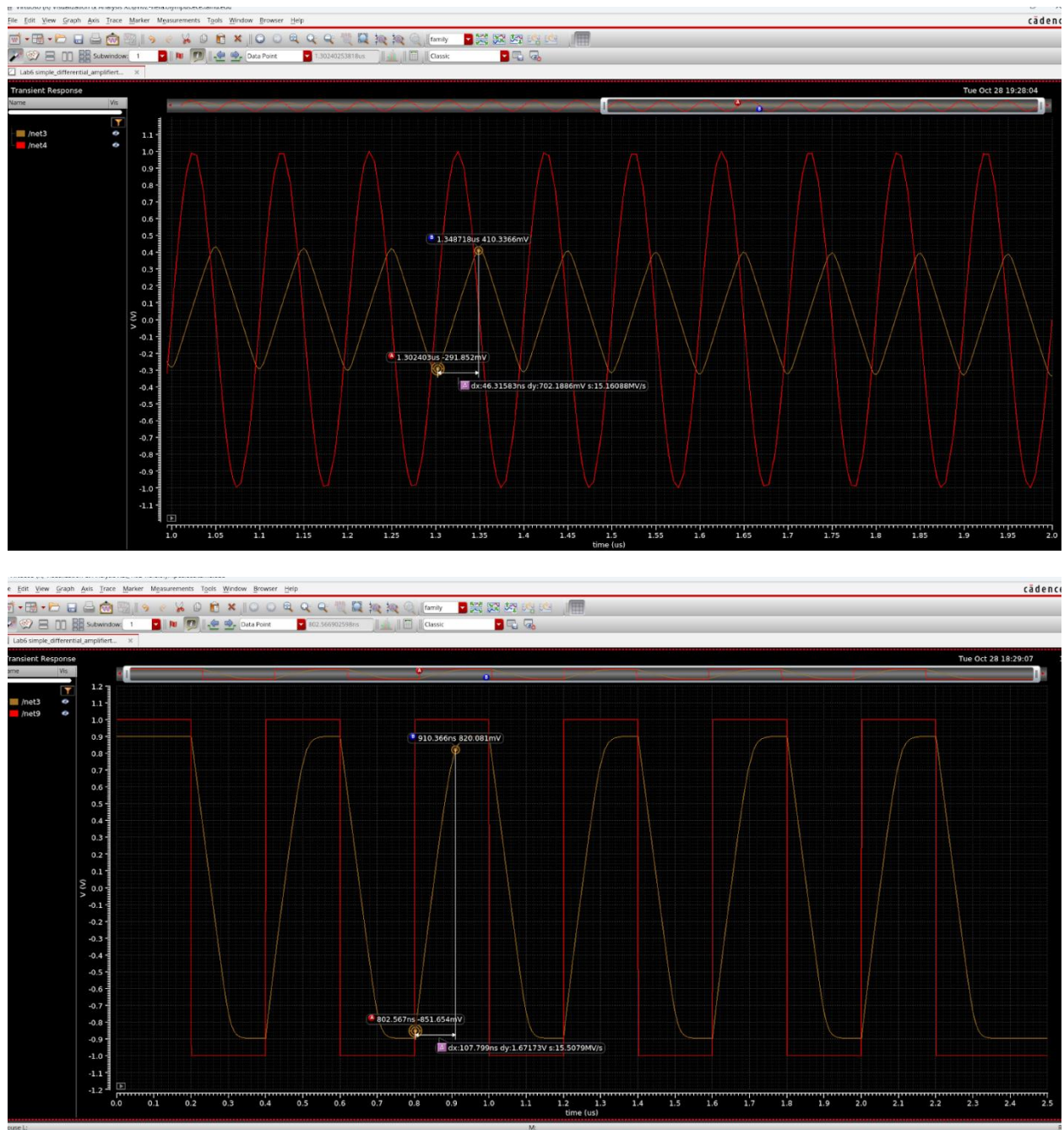
Analysing the waveform of both:



4.CMRR:



5,6,7-SLEW RATE ,GBW and pole frequency:



POST LAYOUT SIMULATION: tried to bring the simulation in single window but faced some issues while performing the simulation:

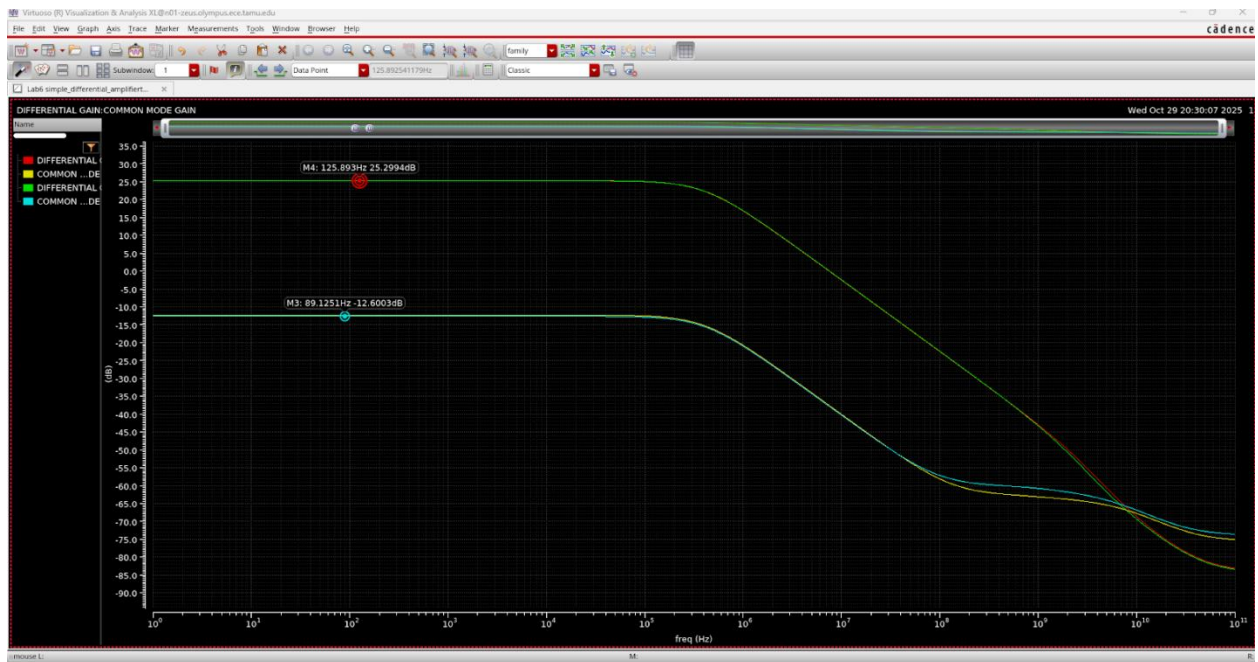
1. Common mode input voltage range, first I analysed the wave taking only the transistors of the output stage for easy analysis since the rest would follow their matched pair:



Analysing with the diff and common mode output voltage sweep:



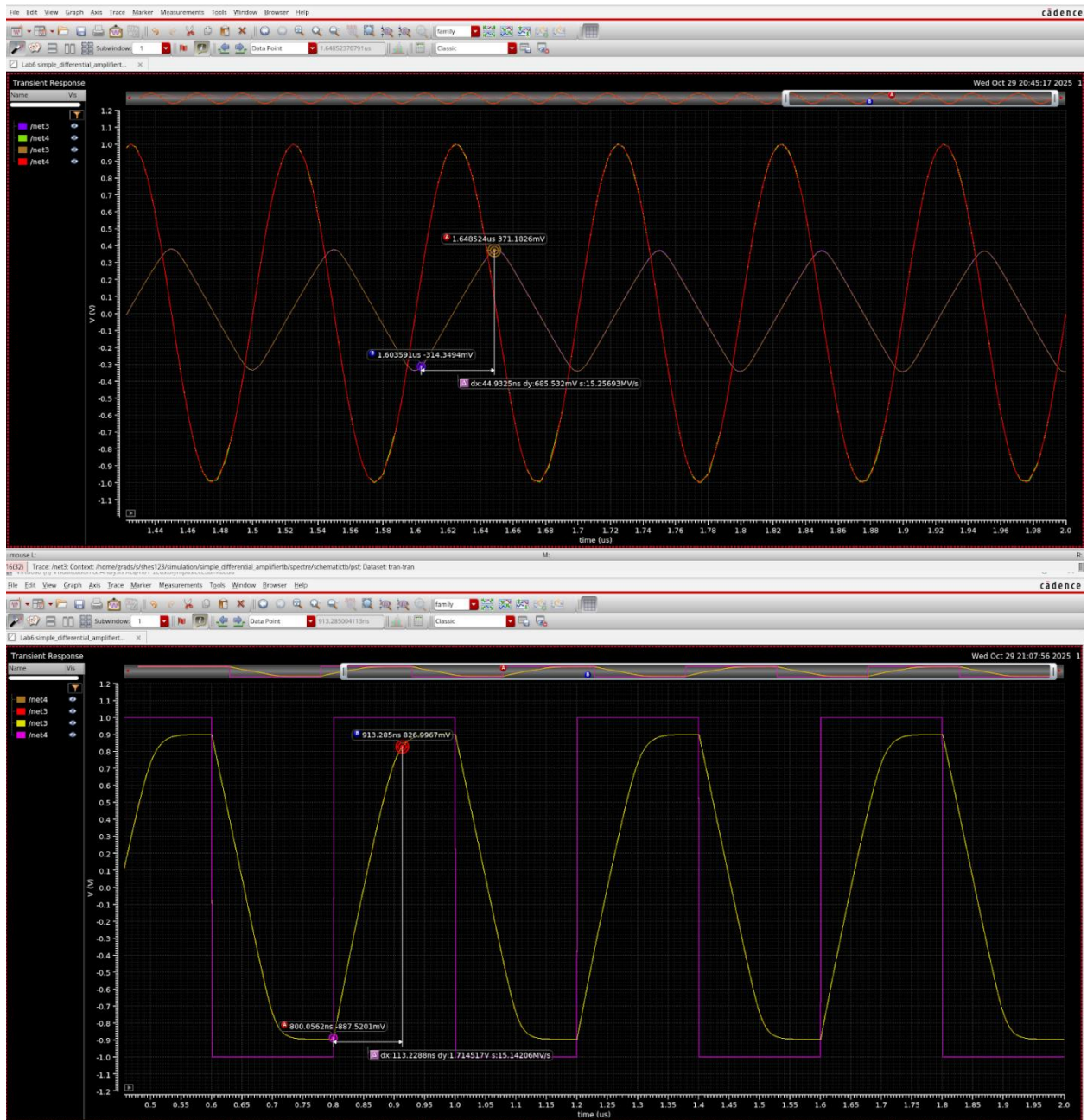
2,3:Common mode gain and differential mode gain:



4. CMRR WITH SLEW RATE POLE FREQUENCY AND GBW VALUES:



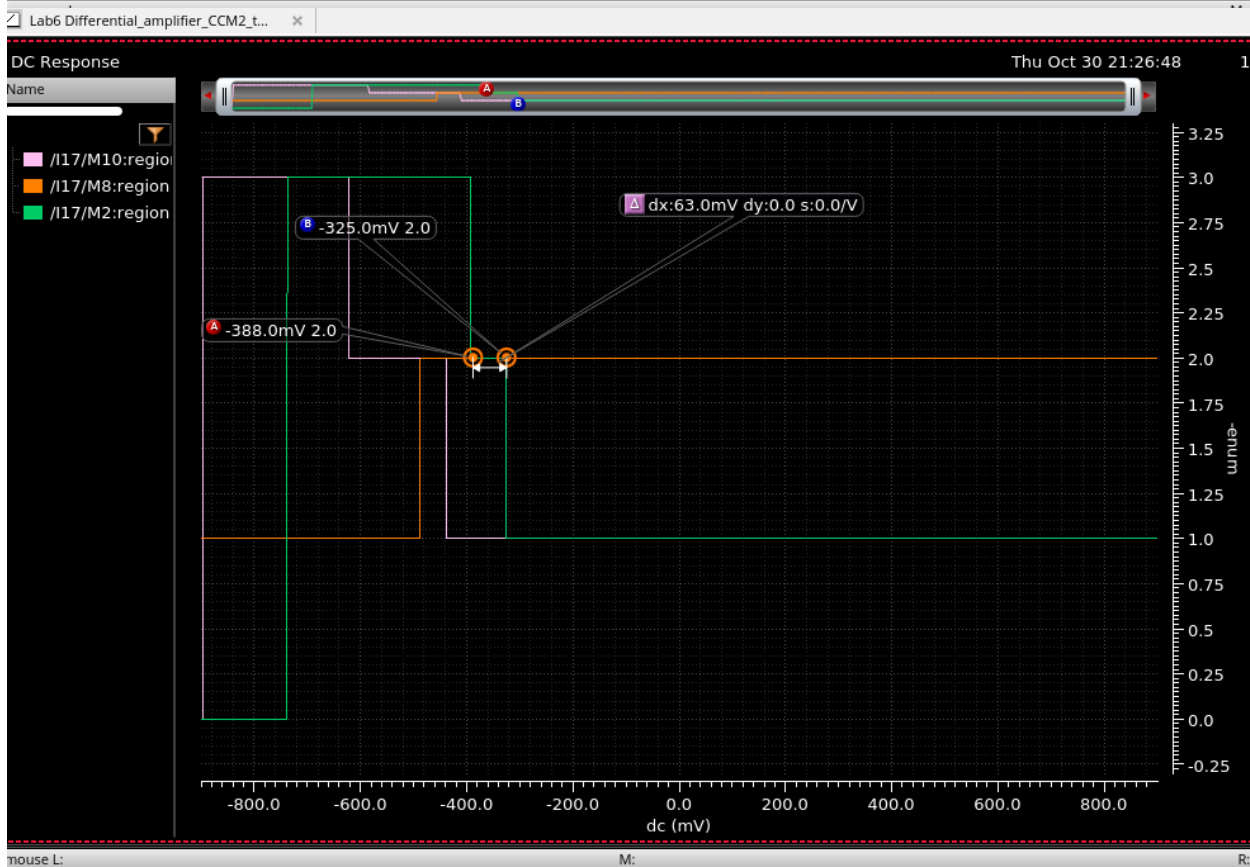
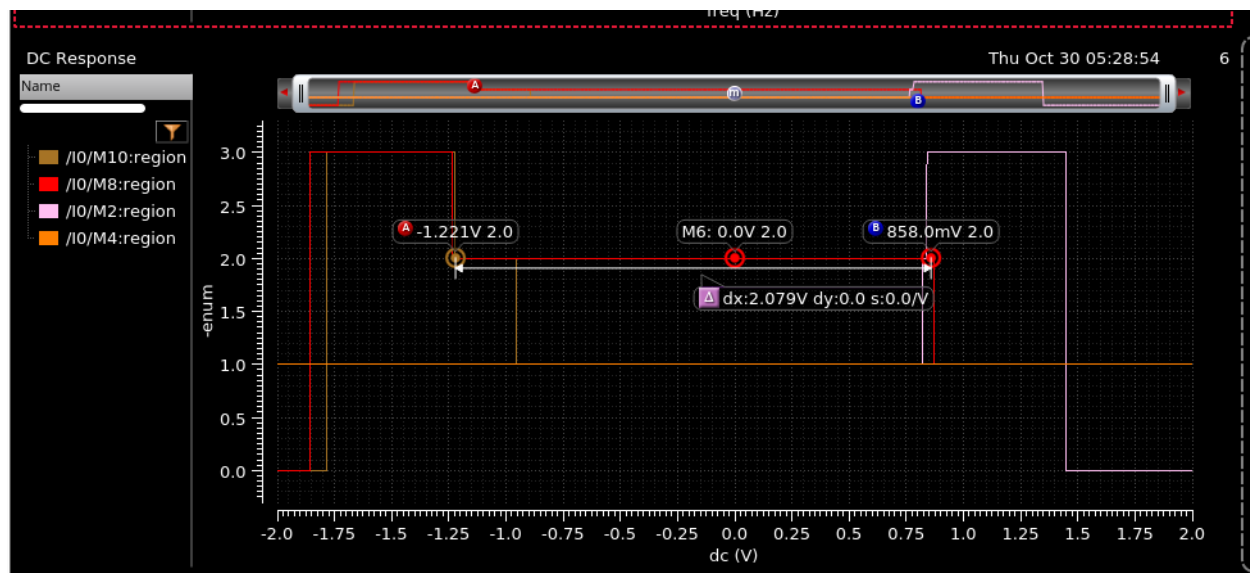
5. SLEW RATE :



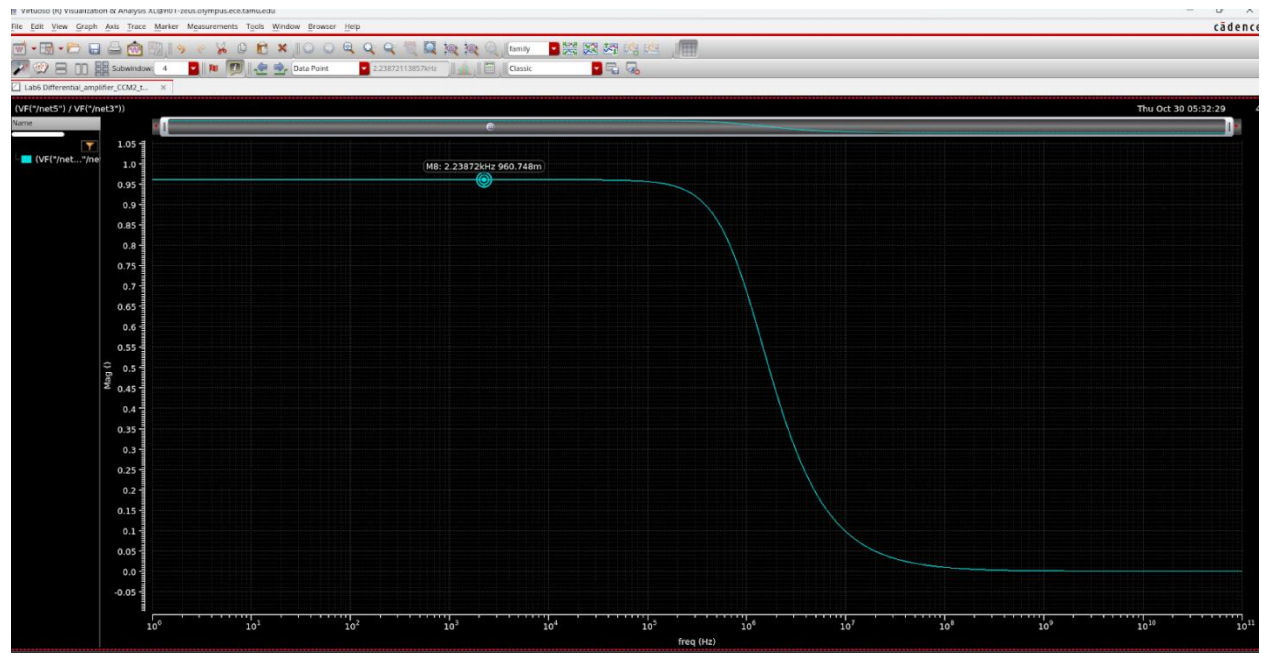
2.CURRENT MIRROR CASCODE DIFFERENTIAL PAIR:

1.COMMON MODE INPUT VOLTAGE RANGE WITH TRANSISTORS OPERATING IN THAT FREQUENCY RANGE:

Continuously troubleshot the values to obtain different graphs at different instances:



OBTAINED MULTIPLE GRAPHS
2.ac plot for differential mode gain:



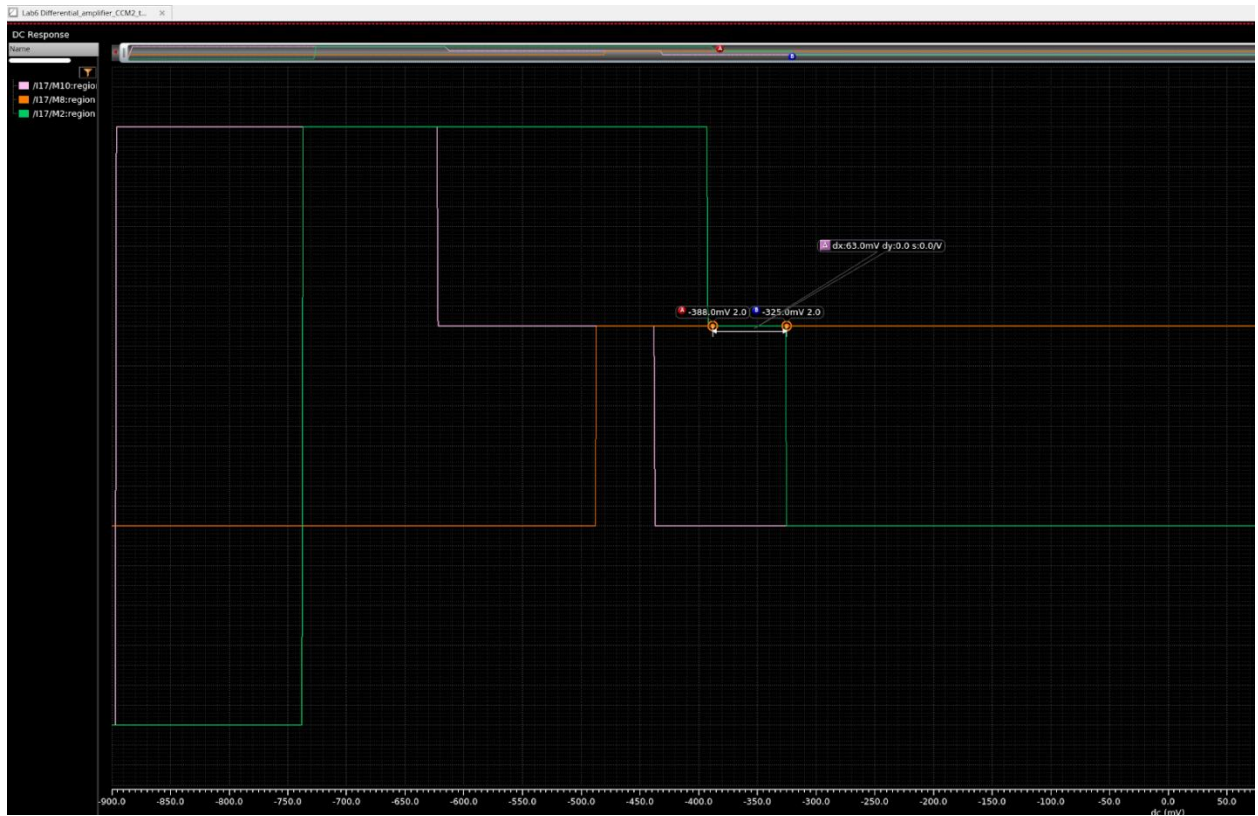
3.ac plot for common mode gain:

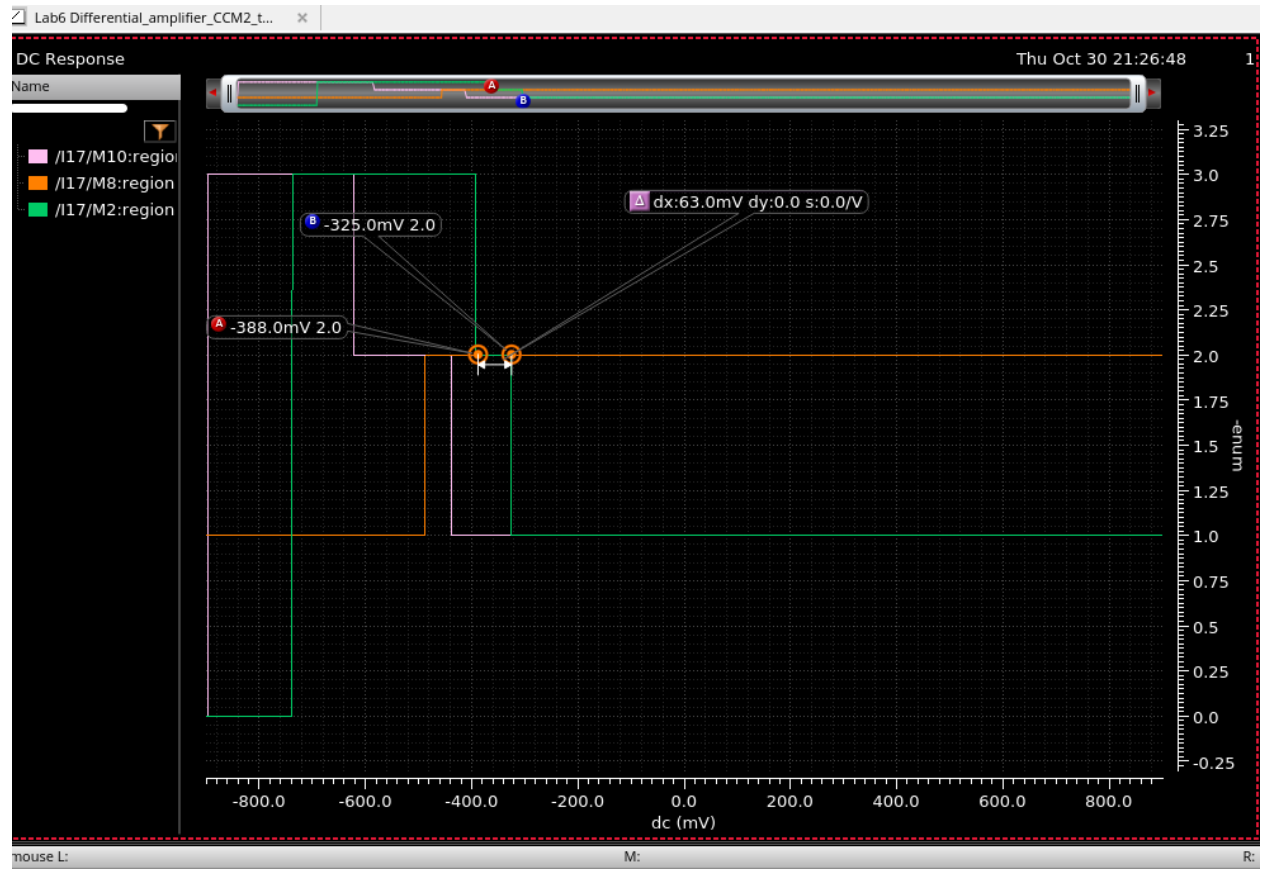


4.CMRR AND RELATED PARAMETERS:

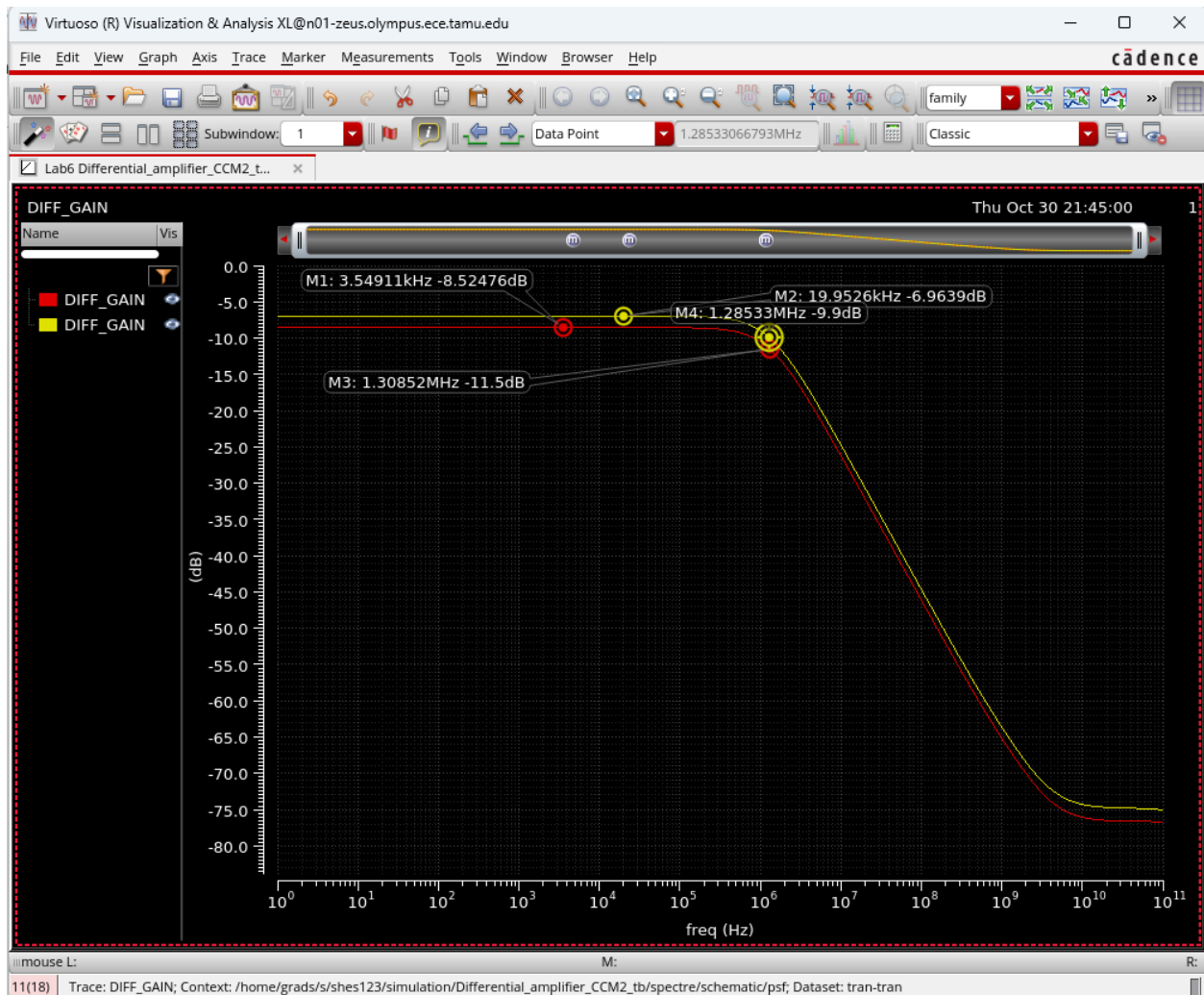


POST LAYOUT SIMULATION:

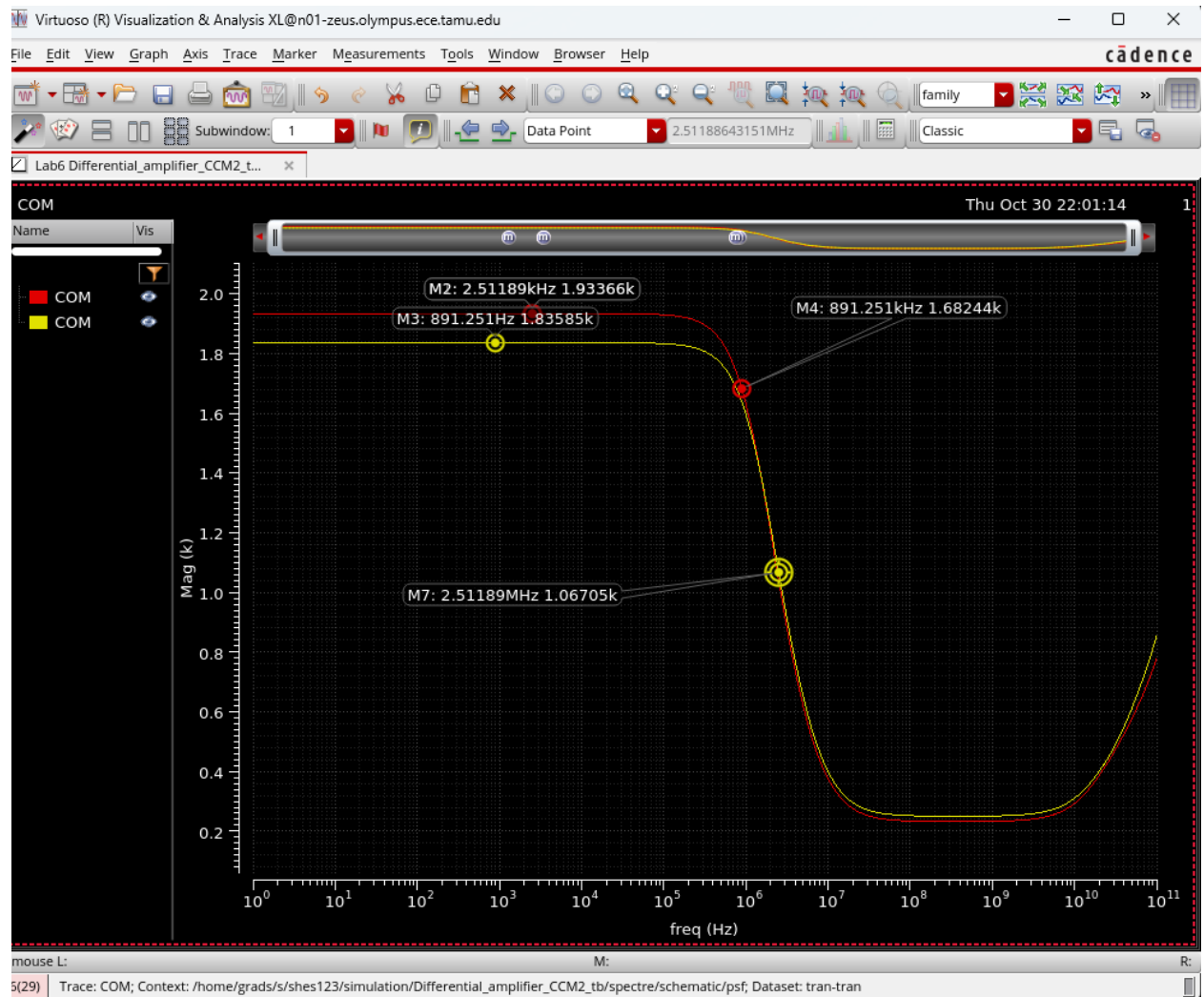




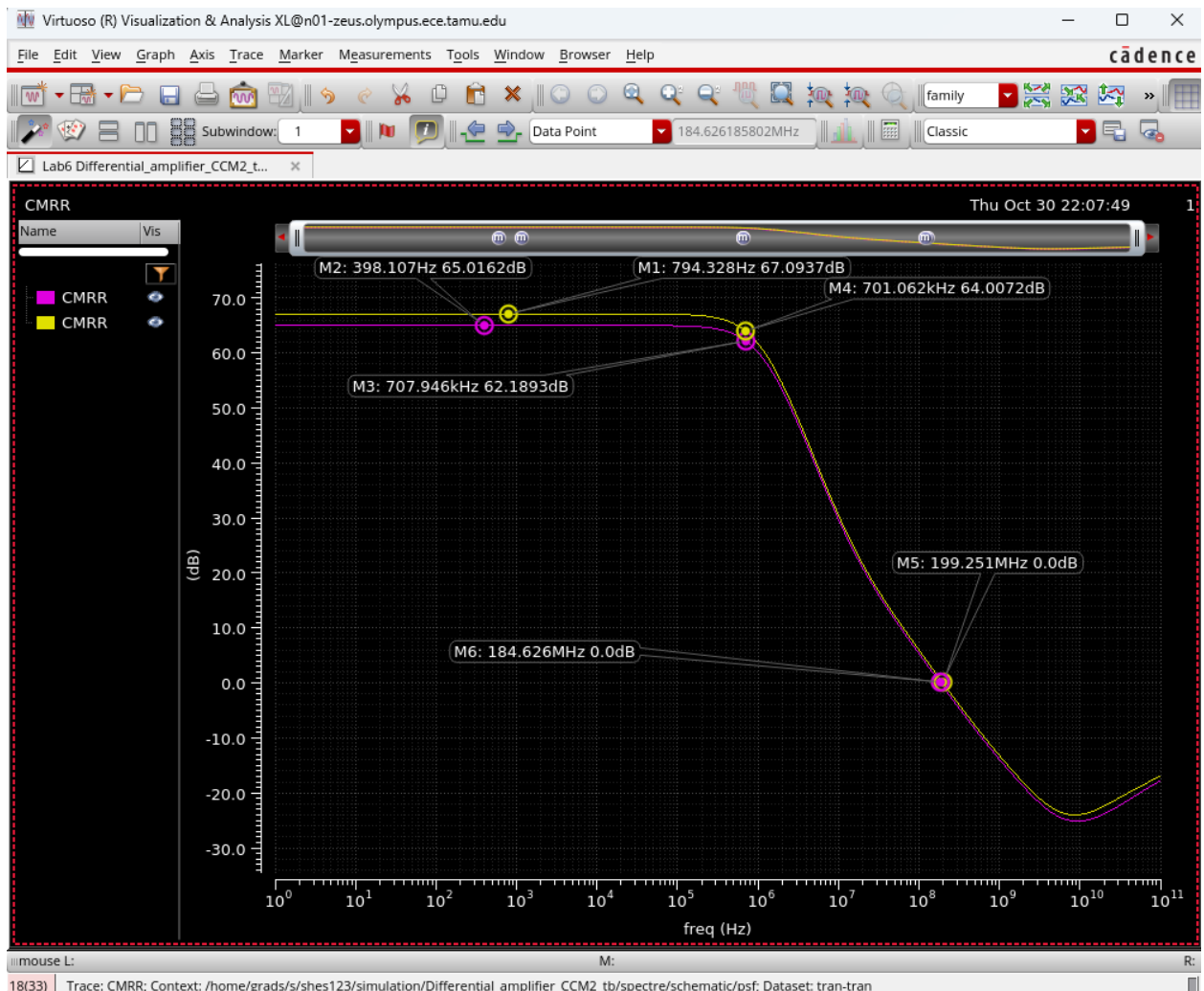
2.DIFF GAIN:



3.COMMON MODE GAIN:



4.CMRR:



SLEW RATE:

